Dr. Mengyan Zhang

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RESEARCH INTERESTS

My research interests are sequential decision-making in machine learning, including Reinforcement learning, Bayesian optimisation and active learning. I work on both theoretical and practical views of experimental design with two goals: (I) Designing robust algorithms to handle imperfect feedback and understand causal relationships in sequential decision-making. (II) Designing decision-making algorithms to solve real-world problems in various areas, for example, synthetic biology, disease surveillance, survey design, and public policy.

WORK & RESEARCH POSITIONS

Lecturer (2025.10-)
School of Computer Science, University of Bristol.

Postdoctoral Researcher (2023.05-2025.09)
Department of Computer Science, University of Oxford.
Affiliated with Kellogg College, Member of Machine Learning and Global Health Network.

Research Consultant (2024.10-2025.09)
Yale University; Project: Predict, Detect, Diagnose: Confronting Outbreaks of HIV and Other Infectious Diseases Among People Who Use Drugs.

NCI-ANU Associate Training Officer (2023.4-2023.7)
NCI Australia. Develop courses and tutorials for machine learning training and cloud service usage.

Research Internship (2021.10-2022.03)
Social Computing Lab, Microsoft Research Asia. Lead, design and deploy algorithms with multi-head attention-based natural language processing (NLP) and contextual bandits algorithms (in PyTorch).

EDUCATION

Ph.D. in Machine Learning
The Australian National University, Data61, CSIRO, 2018.08- 2023.02.
Thesis: Adaptive Recommendations with Bandit Feedback.

Bachelor of Computer Science (first class honours)
The Australian National University, 2014-2018, GPA 6.938/7.0
Thesis: Classification of historical death and occupation coding.

SELECTED AWARDS

2024 Award for Excellence, University of Oxford [top 10%]
2023 CORE Distinguished Dissertation Award Commendation (PhD thesis)
2022-2023 COVID-19 Extension Scholarship (Research)
2018-2022 PhD Scholarship of ANU, Data61 Top-up Postgraduate Research Scholarship
2017 Paul Thistlewatte Memorial Honours Year Scholarship of ANU
2015-2016 National Scholarship (China)

GRANTS

NCCR Automation Fellowship (2024, up to CHF 20,000)
Funded Research Visit at Learning and Adaptive Systems (LAS) Group in ETH Zurich, Switzerland. Work on sequential recommendations and reinforcement learning.

HIGHLIGHT PUBLICATIONS [1]

Artificial intelligence for modelling infectious disease epidemics
*Moritz U. G. Kraemer, Joseph L.-H. Tsui, ..., **Mengyan Zhang** et al (2025). Nature.*

Toward optimal disease surveillance with graph-based Active Learning
Joseph L.-H. Tsuia, **Mengyan Zhang*** et al. (2024). PNAS; KDD 2024 Workshop epiDAMIK*

Quantile Bandits for Best Arms Identification
***Mengyan Zhang**, Cheng Soon Ong. (2021). International Conference on Machine Learning.*

Gaussian Process Bandits with Aggregated Feedback
***Mengyan Zhang**, Russell Tsuchida, Cheng Soon Ong (2022). AAAI.*

Machine Learning Guided Batched Design of a Bacterial Ribosome Binding Site
***Mengyan Zhang** et al. (2022). ACS Synthetic Biology.*

Graph Agnostic Causal Bayesian Optimisation
Sumantrak Mukherjee, **Mengyan Zhang***, Seth Flaxman, Sebastian Josef Vollmer (2024). NeurIPS Bayesian Decision-making and Uncertainty Workshop.*

Opportunities and Challenges in Designing Genomic Sequences
***Mengyan Zhang**, Cheng Soon Ong (2021). ICML Workshop on Computational Biology.*

OTHER PUBLICATIONS/PRE-PRINT

Indirect Query Bayesian Optimization with Integrated Feedback
***Mengyan Zhang** et al. (2024). Under Review.*

Spatiotemporal Inference with Biased Scan Attention Transformer Neural Processes
*Daniel Jenson, Jhonathan Navott, Piotr Gynfelder, **Mengyan Zhang**, et al. (2025). Under Review.*

Uncertainty-Aware Regression via Multi-View Remote Sensing
*Fan Yang, Sahoko Ishida, **Mengyan Zhang**, et al. (2024). Under Review.*

Two-Stage Neural Contextual Bandits for Personalised News Recommendation
***Mengyan Zhang**, et al. (2023). Preprint.*

TECHNICAL SKILLS

- **Programming**: Python, R, Java, C, C++, C#.
- **Machine Learning & Deep Learning:** Proficient in the Python ML stack (PyTorch, Jax, Scikit-Learn, NumPy, SciPy, Pandas, Jupyter Notebooks); Experience in Transformers, BERT, Hugging Face, generative models, Geometric Deep Learning (GNN), LLM.
- **Leadership & Communication**: Led 5+ AI and interdisciplinary projects, mentoring and coaching 10+ team members and students. Experienced in communicating complex technical concepts to policymakers, clients, and cross-sector stakeholders. Completed leadership and innovation training at the University of Oxford (MPLS Division).

[1] *: EQUAL CONTRIBUTION

TEACHING & SUPERVISION

Guest Lecturer (2023.05)
RMIT. Course: Bioinformatics and Multi-omics data analysis (BIOL 2524). Design and deliver 3 lectures on the topics of Introduction to ML and applications in biology for Bioinformatics students. Create original slides and lab materials.

Teaching Assistant (2019-2021)
The Australian National University. Courses: COMP6670 Introduction to Machine Learning; COMP8600 Statistical Machine Learning. Design and deliver material for tutorials and labs. Assess assignments and exams.

Co-supervisor (2023 - 2024)
University of Oxford. Co-supervise 4 students (master's and PhD) in topics of causal decision making, uncertainty quantification, and deep generative models.

College Advisor (2023-2024)
Kellogg College. Provide mentorship to 7 postgraduate students in various subjects in college.

INVITED TALKS & RESEARCH VISIT

Active Disease Surveillance (2024.06)
Machine Learning and Global Health Workshop, Oxford. Co-organisator

Sequential Decision-Making: Theory and Applications in Public Health (2024.02)
LAS Group, ETH Zurich, Switzerland. NCCR Visiting Fellowshop.

Design Choices in Sequential Decision-Making with Bandit Feedback (2023.12)
Causal Intelligence Team, Google DeepMind, London. Visiting Scholar.

Sequential decision making in public health (2023.11)
AIMS seminar, University of Oxford. Invited Speaker.

Bayesian optimisation with aggregated feedback (2023.11)
Bayes@CIRM Workshop, Marseille, France. Invited Speaker.

Sequential Decision-making: Theory and Applications (2023.07)
ADSC Seminar, University of Adelaide. Visiting Scholar.

Bandits in Recommendation System (2022.05)
Social Computing Group Seminar, Microsoft Research Asian. Research Intern and speaker.

REFEREE

Prof. Seth Flaxman
Associate Professor in Department of Computer Science at University of Oxford
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Dr. Cheng Soon Ong
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